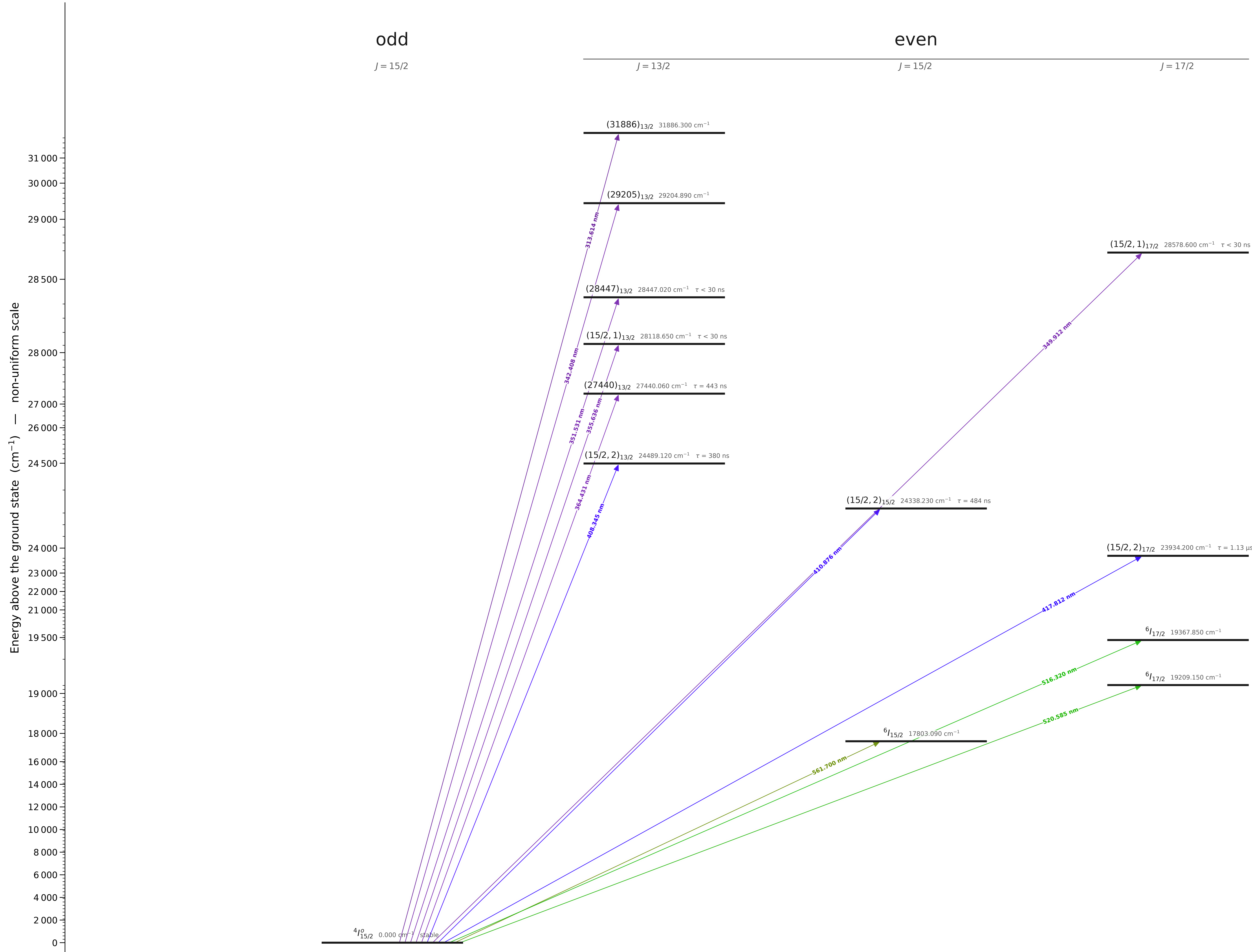


Es (Es I). Classified transitions below 2 μm (levels and lines from the literature): 13 levels, 12 lines.



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e J' \rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^4I_{5/2}^- - (15/2, 1)_{17/2}$	349.9122	349.8121	856.7649	-	-	5.305	-	first step of Mainz resonance ionization scheme No. 10
$^4I_{5/2}^- - (28447)_{13/2}$	351.5307	351.4302	852.8202	-	-	5.305	-	the einsteinium laser-spectroscopy line
$^4I_{5/2}^- - (15/2, 1)_{13/2}$	355.6359	355.5343	842.9759	-	-	5.305	-	first step of Mainz resonance ionization scheme No. 8
$^4I_{5/2}^- - (27440)_{13/2}$	364.4307	364.3269	822.6323	-	-	0.3593	-	Mainz resonance ionization scheme No. 7
$^4I_{5/2}^- - (15/2, 2)_{13/2}$	408.3446	408.2293	734.1653	-	-	0.4188	-	Mainz resonance ionization scheme No. 3
$^4I_{5/2}^- - (15/2, 2)_{15/2}$	410.8762	410.7603	729.6418	-	-	0.3288	-	Mainz resonance ionization scheme No. 2
$^4I_{5/2}^- - (15/2, 2)_{17/2}$	417.8122	417.6944	717.5293	-	-	0.1405	-	Mainz resonance ionization scheme No. 1

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)									
$4_{15/2}^{10}$	817.153(7)	-4316.25(8)	F=4: -25882.65	F=5: -21004.10	F=6: -15343.64	F=7: -9006.96	F=8: -2117.39	F=9: +5184.13	F=10: +12739.06	F=11: +20371.20		
$6_{15/2}^{10}$	2047.400	-2377.400	F=4: -61776.20	F=5: -51102.54	F=6: -38400.88	F=7: -23729.45	F=8: -7156.18	F=9: +11241.31	F=10: +31375.68	F=11: +53149.90		
$6_{17/2}^{10}$	2816.700	-6514.500	F=5: -95930.56	F=6: -77798.63	F=7: -56956.06	F=8: -33546.56	F=9: -7734.36	F=10: +20295.78	F=11: +50338.57	F=12: +82168.20		
$6_{17/2}^{10}$	-113.200	42.000	F=5: +3778.57	F=6: +3091.43	F=7: +2291.77	F=8: +1380.52	F=9: +358.75	F=10: -772.37	F=11: -2011.51	F=12: -3357.20		
$(15/2, 2)_{17/2}$	2284(15)	-	F=5: -75943.00	F=6: -62239.00	F=7: -46251.00	F=8: -27979.00	F=9: -7423.00	F=10: +15417.00	F=11: +40541.00	F=12: +67949.00		
$(15/2, 2)_{15/2}$	1532(15)	-	F=4: -45577.00	F=5: -37917.00	F=6: -28725.00	F=7: -18001.00	F=8: -5745.00	F=9: +8043.00	F=10: +23363.00	F=11: +40215.00		
$(15/2, 2)_{13/2}$	99(15)	-	F=3: -2598.75	F=4: -2202.75	F=5: -1707.75	F=6: -1113.75	F=7: -420.75	F=8: +371.25	F=9: +1262.25	F=10: +2252.25		
$(15/2, 1)_{13/2}$	1303.400	-866.400	F=3: -34547.48	F=4: -29181.55	F=5: -22509.83	F=6: -14556.14	F=7: -5349.03	F=8: +5078.17	F=9: +16687.38	F=10: +29435.75		
$(28447)_{13/2}$	2766(21)	-4425(530)	F=3: -74309.42	F=4: -62467.40	F=5: -47847.22	F=6: -30570.45	F=7: -10782.97	F=8: +11345.03	F=9: +35619.04	F=10: +61820.25		
$(29205)_{13/2}$	1911.500	-2287.400	F=3: -51056.64	F=4: -43008.46	F=5: -33042.50	F=6: -21221.59	F=7: -7621.15	F=8: +7670.85	F=9: +24553.86	F=10: +42914.78		

Measured A, B constants; sources in the data file.

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

$J = 15/2$
 $J = 17/2$

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^4I_{15/2}^o - (28447)_{13/2}$	351.531	254-253	-5020(54)
$^4I_{15/2}^o - (28447)_{13/2}$	351.531	255-253	-11046(86)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$